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Assessing coastal exposure to Sea Level Rise: a coupled approach of qualitative modeling and spatial autocorrelation analysis

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ABSTRACT

Coastal areas are essential for maintaining diverse ecosystems and providing key resources for human populations. However, Climate Change (CC) is a major driver of Sea Level Rise (SLR), manifesting through various impacts such as coastal erosion, flooding, and saline intrusion. This study uses the InVEST model to assess SLR exposure along 25 km of Morocco's Atlantic coastline in the Sale region. The model relies on a qualitative index based on bio-geophysical variables. Four scenarios were analyzed to evaluate the role of natural habitats in coastal protection and how SLR rates influence coastal exposure. The results indicate that with habitat protection, coastal exposure remains between low and moderate (50%), while without protection, 50% of the coastline faces high risk. The impact of SLR further amplifies this risk, with habitat loss resulting in 43.8% of the coastline being classified as "very high risk" and 39.3% as high risk. The southern coastline (Sidi Moussa) is more exposed than the northern part (Nation Beach), due to geomorphology, altitude, distance from the continental shelf, and habitat distribution. The exposure index showed notable spatial autocorrelation (Moran index = 0.7), indicating clustering of areas with similar exposure. The Moran index remained consistent across all scenarios, highlighting stable spatial dependency patterns. These findings help identify high risk districts along the Sale coastline, providing valuable support for coastal protection and sustainable development.

1. Introduction

The coastal zone is the transitional zone between the continent and the sea. This area is known for its specific natural resources, ecosystems, climates, and complex hydrodynamic characteristics (*Rangel-Buitrago and Ben-Haddad, 2024*). These areas offer an attractive living environment due to the variety of ecosystem services and their exceptional ecological and biological productivity (*Maanan et al., 2018; Rizzo et al., 2020*). Consequently, the region is facing major challenges associated with Climate Change (CC), intensifying the vulnerability of these ecosystems and contributing to Sea Level Rise (SLR) (*Mahapatra et al., 2015*).

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SLR is considered one of the primary hazards to coastal communities and environments (Khan et al., 2020; Mills et al., 2020). This can be reflected in the increased frequency of storms and disasters such as submersion, coastal erosion, deformation of coastline morphology, and marine intrusion (Michener et al., 1997; Zscheischler et al., 2018). The Sixth Assessment Report from the Intergovernmental Panel on Climate Change (IPCC) highlights the acceleration of SLR in the coming years, projecting an increase of 2 m by the end of the century under the Shared Socio-economic Pathways (SSP) scenarios (“AR6 Synthesis Report,” n.d.).

Morocco has a 3500 km coastline stretching between the Atlantic Ocean and the Mediterranean Sea (Snoussi et al., 2008, 2010). However, this coastline is subject to various pressures, including human activities, environmental challenges, and climatic pressures, such as dune erosion, flooding, or pollution, which are intensifying due to the effects of SLR (Abdelmounim El Mrini et al., 2012). Recently, Morocco has shown a growing interest in coastal areas research and a deep reflection on its legislation for coastline protection (the application of law 81.12 relating to the littoral). Many studies have evaluated the vulnerability of the Moroccan coastline to various hazards caused by SLR. These studies employ diverse methodological approaches to analyze the phenomena. One of the most common methods is geospatial modeling, which maps shoreline evolution and estimates erosion hazards (Nmiss et al., 2022; Erraji Chahid et al., 2024). Another key approach is hydrodynamic modeling, which is used to evaluate sediment transport scenarios and simulate the response of beaches under the effects of SLR (El Mrini et al., 2012). In addition, some studies use Coastal Vulnerability Indices (CVI) that integrate physical and socio-economic data to assess the impact of coastal hazards (Satta et al., 2016; Fannassi et al., 2023). Statistical analyses and predictive models for anticipating coastal flooding and its potential consequences (Hakkou et al., 2019; Chtioui et al., 2024). Finally, certain studies have focused on the management of coastal aquifers, examining the effects of saltwater intrusion caused by SLR (Ketabchi et al., 2016).

The Sale coastline is vulnerable to various SLR related hazards, including marine submersion, coastal erosion, and the salinization of the Bouregreg estuary. These risks will have impacts on ecosystems, water resources, agriculture, the population, and the regional economy (“bk. EFH-MO-5 (2019)). Concerning the marine submersion hazard, a surcharge of +2.65 m would result in partial or total submersion of all Rabat and Sale beaches (“bk. EFH-MO-5 (2019)).

Various models have been developed to assess coastal vulnerability to SLR based on geophysical attributes. These include the Coastal City Flood Vulnerability Index (CCFVI) (Balica et al., 2012), the Coastal Vulnerability Index (CVI) (Gornitz, 1990), and Multi-Scale CVI (McLaughlin and Cooper, 2010). And other models, such as the CVI (SRL) (Özyurt* and Ergin, 2010), the Social Vulnerability Index SoVI (Guillard-Gonçalves et al., 2015), and the Composite Vulnerability Index (Szlafsztein and Sterr, 2007), as well as others that qualify the role of the habitat in protecting coasts against SLR like Delft3D model (Wiyono et al., 2024) and SLAMM (De Souza De Lima et al., 2024). In particular, InVEST (Integrated Valuation of Environmental Services and Trade-offs) provides assessments and mapping of exposure of Sale coastline to coastal hazards, guiding actions to improve adaptability for sustainable development. This model is based on a qualitative index incorporating biological, geophysical, and climatic parameters. The InVEST coastal vulnerability model integrates ecological, social, and economic elements, offering a complete analysis. Its spatially explicit outputs generate detailed vulnerability maps, facilitating the identification of specific high-risk segments. The model provides scenario-based analysis, crucial for evaluating the potential impacts of CC or management strategies. Furthermore, InVEST is user-friendly and open-source, making it available to many users. Its proven efficacy in usage confirms its dependability as a tool for coastal management (Arkema et al., 2013a; Sharp, 2023) the advantage of using this model is that it combines these two objectives, assesses the exposure of the coastline to SLR, and highlights the importance of habitat management in reducing the risk to the coastline.

This research is the first to utilize the InVEST coastal vulnerability model to evaluate coastal exposure

to hazards in Morocco. This approach differs from other methods by evaluating exposure under various scenarios and highlighting the significant role of natural habitat in reducing risk. To our knowledge, this is the first time this approach has been applied in research conducted in Morocco, and it also includes an analysis of geospatial autocorrelation.

The study aims to evaluate the Sale coastline's exposure to SLR using the InVEST coastal vulnerability model. It specifically highlights the role of natural habitats in mitigating risks. It uses spatial autocorrelation analysis to understand exposure's spatial distribution. The aim is to provide insights for coastal management and guide sustainable development strategies to enhance resilience against SLR impacts.

This study combined an approach using the InVEST model and spatial autocorrelation analysis (global and local) (Ai *et al.*, 2022). The results were validated through a field investigation based on findings from the first scenario. The methodology begins with data collection and processing, then classifying relevant parameters. Next, the coastal Exposure Index (EI) is calculated using the InVEST model, and the results are mapped to create a coastal EI map. A spatial autocorrelation analysis of the Exposure Index uses the global Moran index and Local Indicators of Spatial Association (LISA) to identify spatial patterns. Finally, field surveys are conducted to validate the results through a comparison with the first scenario and on-site observations of already eroded areas.

2. Material and methods

2.1 Study area

The coastal region of Sale is situated in northern Atlantic Morocco on latitude 34° W, Longitude -6.8° N. It is part of the Rabat-Sale-Kenitra region in north-eastern Morocco (**Figure 1**). Geographically, the prefecture of Sale is located on the right bank of the mouth of the Bouregreg estuary and between the province of Kenitra to the north, Rabat to the south, and the Atlantic Ocean to the west (**Figure 1**). The geomorphology is characterized by three distinct morphologies (Aberkan, 1989; Hakkou, 2012): a sandy beach to the north of Sale (nation beach) that extends to Mehdiya with high dunes, cliffs with prominent formations at Sidi Moussa and Saïd Hajji, and Bouregreg estuary (**Figure 3**).

The climate in this region is Mediterranean with an oceanic influence and belongs to the sub-humid bioclimatic zone. The nearby Maâmoura forest significantly affects its microclimate (Elhachimi *et al.*, 2022). The average annual rainfall is 540 mm, ranging from 300 mm to 900 mm. The rainy season occurs from October to March, while the dry season is from April to September. The prevailing winds are mostly westerly during the summer (56%) and easterly during the winter (25%), with calm winds accounting for 19%. These winds contribute to an estimated potential flow of around 20m³/ml/year to the land (Elhachimi *et al.*, 2022).

The hydrodynamic regime is subject to long swells, the influence of which is felt above all in the coastal surf zone, which is around 400 to 500 m wide. The dominant swells come from the north-west (300°N to 320°N) with typical periods of 10 to 15 seconds, 70% of which have a significant height of between 0.5 and 2m (Hakkou *et al.*, 2011; Mhammedi *et al.*, 2020). Swells are crucial for coastal sediment transport, especially those that arrive obliquely to the shoreline, generating parallel longshore currents (Khalid *et al.*, 2009). Long-period swells, accounting for 60%, are the most effective in mobilizing and transporting sediments over long distances (Khalid *et al.*, 2009). Powerful swells, occurring two or three times a year, reach heights of 7 to 9 meters, while powerful swells, exceeding 9 meters, occur every two or three years (Chahid *et al.*, 2016). The tides are mesotidal (range 1-3.4 m) and semi-diurnal (12h 24 min).

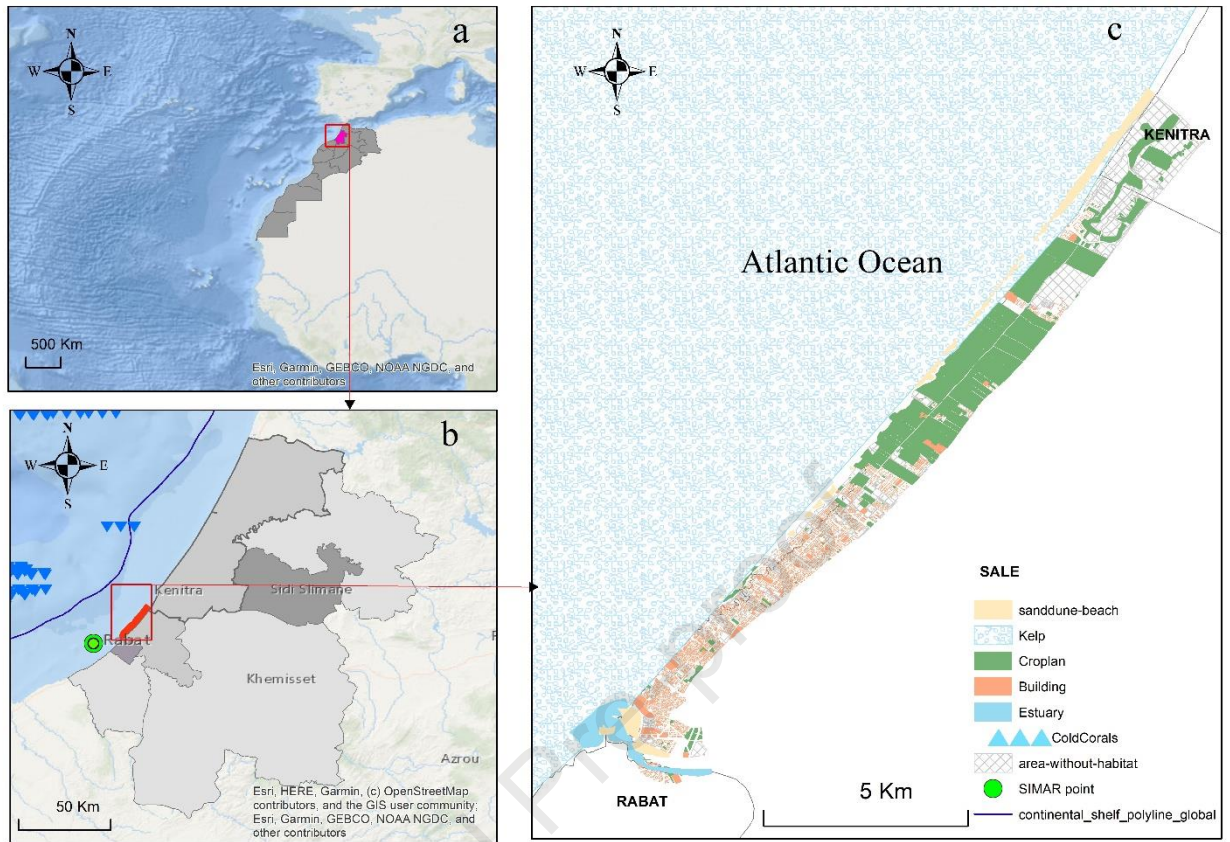


Figure 1: location of the study area, a) at the world scale, b) at the regional scale, and c) at the local scale.

2.2 Methodology

This research employs a methodological approach using a geographic information system (GIS) to evaluate coastal exposure. **Figure 2** summarizes the approach used in this study. In step 1, the input data are processed and classified using a GIS based on the guidelines proposed by the InVEST model package. In step 2, the model is run to compute the Exposure Index (EI) using the input factors. Step 3 consists of mapping the output and analyzing the degree of coastal exposure. Step 4, examines the spatial autocorrelation (global and local) of the parameters influencing the index. This provides an overall view of the factors contributing to spatial variations of EI.

To validate the results of the InVEST coastal vulnerability model, a visual analysis was conducted using images collected from the study area. These photos were used to support the model's identification of the exposure of the coastline results, by comparing real observations with the current scenario (present condition) of the expected regions. The photos were geolocated and placed on the model's maps created using a Geographic Information System (GIS). Visible indicators, such as erosion, debris accumulation, and exposed infrastructure, were compared to EI results.

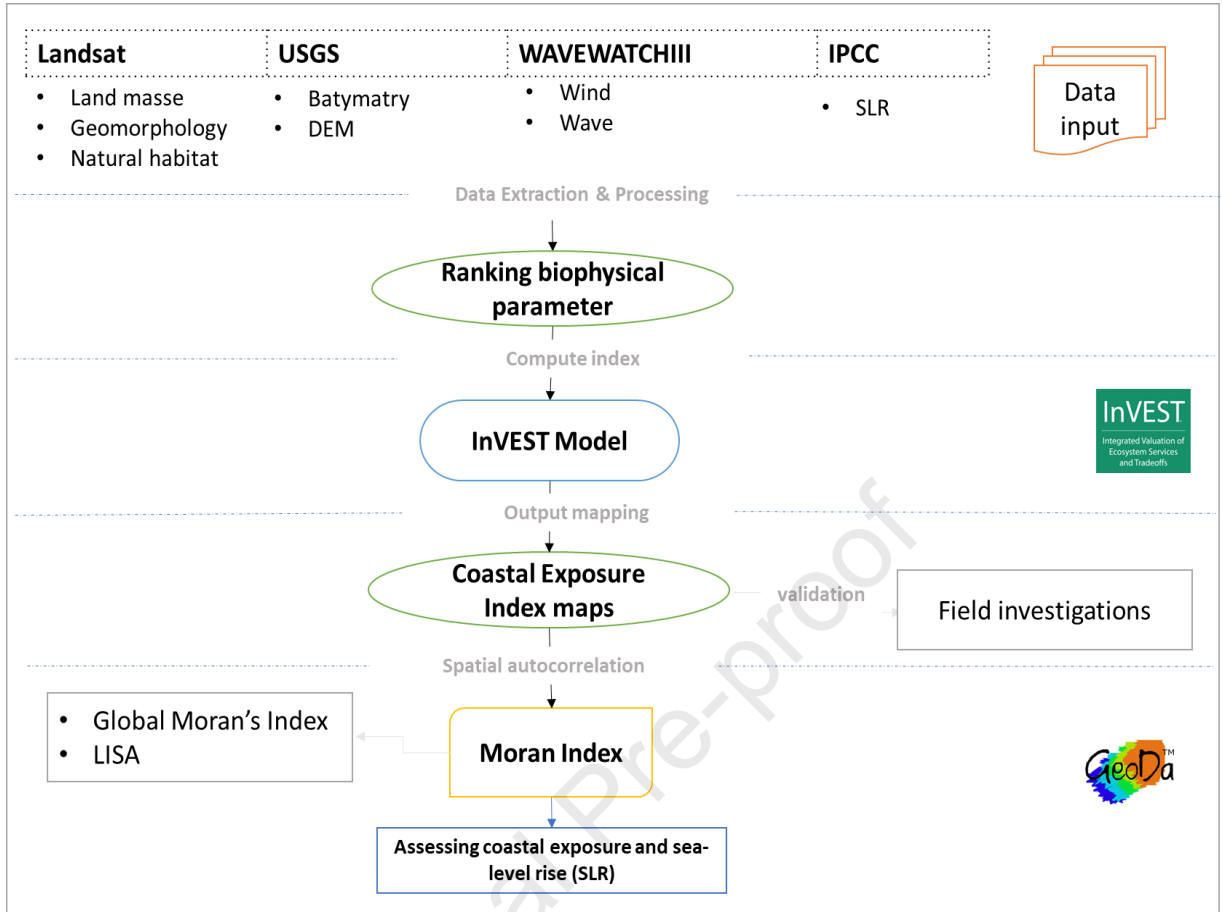


Figure 2: The flow diagram of the Coastal Exposure Used in this research.

The Exposure Index (EI) was calculated using the open-source InVEST, version 3.13.0 (Sharp, 2023) developed by the Natural Capital Project. The EI represents the relative exposure of different coastline segments to erosion and inundation caused by storms within the region of interest. The InVEST model is widely recognized as an effective tool for assessing coastal exposure, particularly in regions with limited data availability (Arkema et al., 2013a, 2017; Hopper and Meixler, 2016; Cabral et al., 2017; Sharp, 2023). The InVEST model is one of many previously proven approaches for assessing the coastal exposure of each shore segment within the domain of interest relative to all other shore segments. It incorporates physical exposure factors and the protective effects of natural habitats while considering the influence of local management actions. The model computes the coastal exposure index by combining up to seven biological and physical variables at each shoreline point (geomorphology, natural habitats, relief, wave height, wind exposure, surge, and sea-level rise) at various coastal locations. These variables are assigned rankings on a scale from **1 (very low exposure)** to **5 (very high exposure)** based on their influence on coastal exposure. The ranking method follows the approaches proposed by (Gornitz, 1990; Hammar-Klose and Thieler, 2001). This study used the model's default resolution of 100 meters, resulting in 267 points along the coastline. The Exposure Index is then calculated by taking the geometric mean of the rankings of these seven variables, as shown in the following **Equation 1**.

$$EI = (R_{geomorphology} \cdot R_{natural} \cdot R_{relief} \cdot R_{wave} \cdot R_{wind} \cdot R_{surge} \cdot R_{SLR})^{1/7} \text{ (Equation 1)}$$

R represents the ranking of each biophysical variable; R geomorphology refers to the rank of geomorphology for each coastline, R natural denotes the rank for natural habitats, R relief indicates the ranking of relief, R

wave corresponds to the average height of adjacent seas, R wind represents the wind exposure ranking, R surge measures the distance between the coastline and the edge of the continental margin, and R_{SLR} refers to Sea Level Rise.

The model results are presented in thematic maps, where each coastal segment is assigned a level of exposure according to its ranking of EI. These maps illustrate the variation in coastal exposure between different segments. EI classification methods vary from study to study, with the literature reporting no standardized approach to assessing coastal vulnerability (Onat et al., 2018). Some studies apply national scores specific to their study areas (Al Ruheili and Boluwade, 2023; Onat et al., 2018), while others use quartiles divided into three categories (Arkema et al., 2013a; Silver et al., 2019; Manes et al., 2023). Others used the quantile method (Cabral et al., 2017; Lopes et al., 2022). In this study, we implemented a consistent EI classification across all scenarios, our approach after (Cunha et al., 2021) work in northern Portugal, (Jackson et al., 2020) in the Gulf of Mexico, and (Zhang et al., 2020a, 2021; Aja et al., 2023) in China. They categorize exposure into five distinct levels: **Very Low (≤ 1)**, **Low (1-2)**, **Moderate (2-3)**, **High (3-4)**, and **Very High (≥ 4)**. This methodology facilitates a more harmonious comparison of exposure levels and their variation along the coastline in various scenarios.

2.3 Data input processing

Coastal elevation is crucial for predicting and estimating the land area that could be at risk from coastal hazards due to potential storm surges and rising sea levels (Kumar et al., 2010; Murali et al., 2013). Relief is the average elevation of the coastline, measured from each shore segment of the discretized shoreline within a radius (default = 5 km) proposed by the InVEST package (Natural Capital Project 2023). The model recommended a ranking standard (of 1 to 5) that takes into consideration that high-altitude areas are thought to be less vulnerable compared to low-lying areas that are thought to be extremely vulnerable (Hoque et al., 2018).

Coastal geomorphology significantly influences how the shoreline responds to various hazards (Arun Kumar and Kunte, 2012; Islam et al., 2015). The geomorphological characteristics indicate the coastal segments' ability to resist erosion and inundation hazards. The objective was to identify the main types of shorelines and highlight their varying susceptibilities to erosion. The geomorphological map was created in the GIS environment (**Figure 3**). The polyline layer consists of nine geomorphology classes (Table 3): Sand dune, cliff/fjord, Low cliff, Small_ seawalls, seawalls, harbour, and Estuary. The inputs were classified and ranked with a rating from 1 (extremely low erodibility) to 5 (extremely high erodibility), based on earlier research (Gornitz, 1990; Hammar-Klose and Thieler, 2001; Zhang et al., 2021, 2020; Thakur et al., 2021).

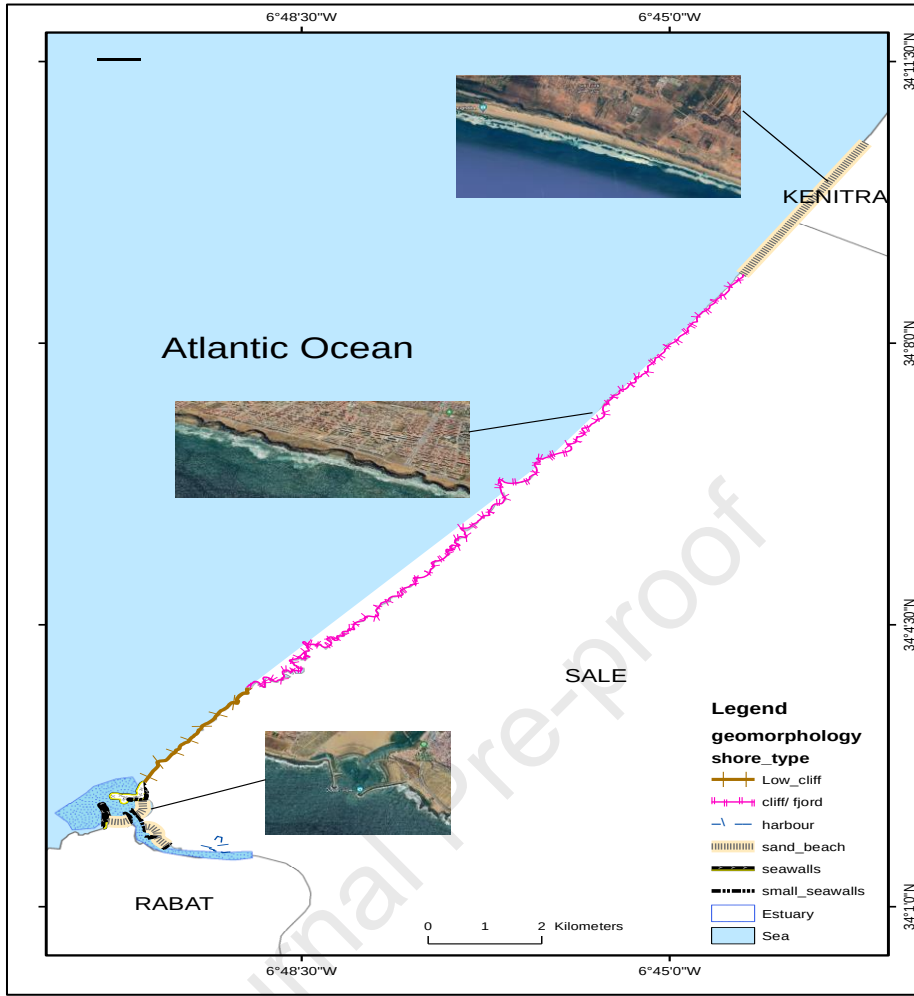


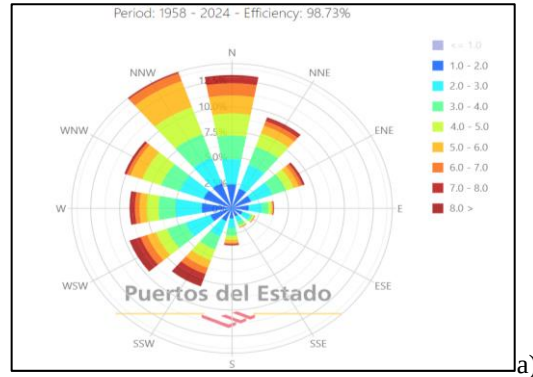
Figure 3: The coastline geomorphology map.

Coastal habitats are essential for mitigating the effects of coastal hazards on local communities (KOBAYASHI *et al.*, 1993; Giri *et al.*, 2011; Ferrario *et al.*, 2014; Bunting *et al.*, 2018; Sajjad *et al.*, 2020a; Chen, 2021). Spatial habitat distribution along the coastline has been determined from land use considering a distance of 1 to 2 km from the shoreline (Figure 1). There are seven major habitats, each with protection distance and exposure ranking (Table 1). The habitat rankings shown in (Table 3) are determined from qualified expertise and reviewed literature (Shepard *et al.*, 2011; Arkema *et al.*, 2013; Spalding *et al.*, 2014; Narayan *et al.*, 2016; Sharp, 2023), which natural habitats influence the maximum protective distance for the shoreline segment, ranges from "1" to "4", with "5" signifying no protection (Table 1). To add this variable to the InVEST model, a habitat table (CSV) was provided to guide the model on the habitat inputs (Table 2).

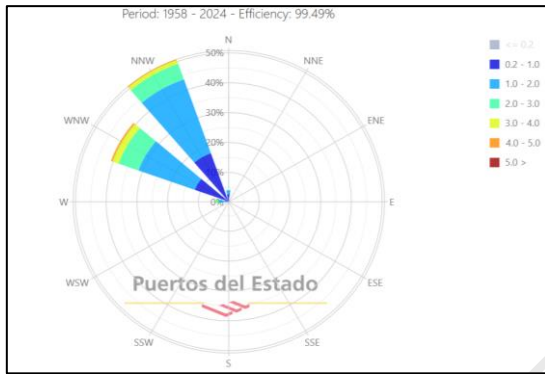
Table 1: Natural habitat of area study and protective distances.

Id	Rank	Protection distance (m)	Source
Building	1	1000	(Hopper and Meixler, 2016; Ai et al., 2022)
Sand dune-Beach	3	100	(Onat et al., 2018; Silver et al., 2019; Lopes et al., 2022; Manes et al., 2023)
Cropland	2	500	(Hopper and Meixler, 2016 ;Ai et al., 2022;)
Seagrass	4	5000	(Hopper and Meixler, 2016; Cabral et al., 2017; alruheili, 2022; Lopes et al., 2022; Manes et al., 2023)
Coldcorals	1	2000	(Silver et al., 2019; alruheili, 2022; Lopes et al., 2022; Manes et al., 2023)
Kelp	4	4000	(Hopper and Meixler, 2016)
Area-without-habitat	5	70	(Onat et al., 2018; Silver et al., 2019)

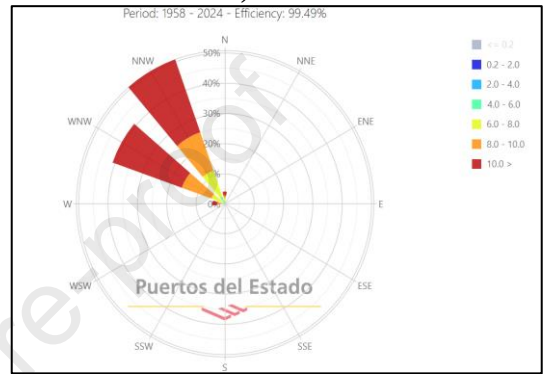
The Relative Exposure Index (REI) (Keddy, 1982) ranks shoreline segments based on their exposure to strong winds. The compass rose (**Figure 4**), also known as the 360-degree compass, is divided into 16 equi-angular sectors, using the highest 10% wind speeds from a long record. The model calculates the fetch distance for a specific shore point by projecting rays in 16 different directions and determining the maximum length of each ray before its intersection with a landmass. The Wave Watch III dataset offers data on wind speed and direction for the Wave Exposure variable used in the InVEST Model. The percentage of a coastline exposed to storm waves is a qualitative indicator of coastal erosion. The strongest waves (>8.0 m) come primarily from the NNW and WNW directions (**Figure 4**). the InVEST download package default wind and wave data were compiled from 8 years of WAVEWATCH III (WW3, Tolman; 2009). The model calculates the relative exposure of a location to waves (Irish and Resio, 2010). The elevation of storm surges is determined by wind direction, speed, and duration over shallow areas. The model calculates the relative exposure to storm surges at every shore point using bathymetric data collected from USGS at a spatial resolution of 30m.



a)



b)



c)

Figure 4: (a) Rose of mean speed (m/s), (b) rose of significant height(m) , and (c) rose of Peak period(s) for SIMAR point 1052036 (Longitude 7.00° W Latitude 34.00° N)

The model can incorporate SLR as a seventh variable in the EI calculation. To our knowledge, no local tide gauge data is currently available. The input is based on the SLR's future projections based on the SSP5-8.5 scenario of the IPCC (*"AR6 Synthesis Report," n.d.*), representing the worst possible case that anticipates the greatest increases in sea level by the end of the century (1.5 m for 2100). The projected SLR values are categorized into five ranks (**Table 3**) (Silver *et al.*, 2019). The raster elevation dataset was created by mapping the high estimate (1.5 m) in GIS using Map Algebra tools (Lopes *et al.*, 2022). The model then required the conversion of this raster dataset into a vectorial format, representing the elevation values at discrete points.

Tables 2 and 3 present the data used in the InVEST Coastal Vulnerability model, which includes variables such as geomorphology, relief, bathymetry, wind and wave exposure, coastal habitat management, and SLR rate. The table highlights the data sources, formats, and resolutions, ensuring transparency and providing a solid foundation for interpreting the results.

Table 2: Data input into the invest coastal vulnerability model for calculating the EI.

Variables	Datasets	Collection	Description	Source
Relief	Relief	Digital Elevation Model (DEM) (ALOS PALSAR RTC with 12.5 m spatial resolution)	The relief ranking of each shoreline segment was calculated using the average elevation of the coastline measured from each shore segment within a radius of 5 km proposed by the InVEST package.	https://asf.alaska.edu/data-sets/sar-data-sets/alos-palsar/alos-palsar-coverage-maps/
Geomorphology	Land cover	Digitalization from Google Earth 2023	Lines were created to represent the different features visible on the Google Earth satellite image of the Sale coastline.	Google Earth
Natural habitats	Land Cover	The digital data were extracted from Google Earth 2023. In addition, data on seagrass, cold corals, and kelp were downloaded from the Ocean Data Viewer platform.	Files were compiled by creating polygons from a variety of sources that mapped the distribution of coastal and ocean habitats along the Sale coast	Google Earth & https://data.unep-wcmc.org/
Wind	WA-VEWATCHIII	Wind speed/h from InVEST model	To estimate fetch distance for a given shore point	https://naturalcapitalproject.stanford.edu/software/invest/invest-downloads-data
Wave	WA-VEWATCHIII	Wave speed/h from InVEST model	The model estimates the relative exposure of a shoreline point to waves	
Surge	Bathymetry	from USGS at a spatial resolution of 30m	This input is used to compute the average depth along the fetch rays to determine each shoreline segment's exposure and the surge potential computation.	https://download.gebco.net/
	Continental shelf	From InVEST model	Continental shelf Depth contour used as an indicator for surge level.	https://naturalcapitalproject.stanford.edu/software/invest/invest-downloads-data
SLR	Sea level change	IPCC SSP5-8.5 scenario	Created point shape file with attributes of sea level trends	https://sealevel.nasa.gov/ipcc-ar6-sea-level-projection-tool?Psm1_id=936

Table 3: Ranking of the variables of coastal EI from (Arkema et al., 2013a; sharp, 2023)

variables	Rank				
	1 (very low)	2 (low)	3 (moderate)	4 (high)	5 (very high)
Geomorphology	high cliffs;	small seawalls	Low cliff	estuary	sand beach, harbor
Relief	81 to 100 Percentile	61 to 80 Percentile	41 to 60 Percentile	21 to 40 Percentile	0 to 20 Percentile
Natural Habitats	Coral reef; mangrove; coastal forest	High dune; marsh	Low dune	Seagrass; kelp	No habitat
Wave Exposure	0 to 20 Percentile	21 to 40 Percentile	41 to 60 Percentile	61 to 80 Percentile	81 to 100 Percentile
Surge Potential	0 to 20 Percentile	21 to 40 Percentile	41 to 60 Percentile	61 to 80 Percentile	81 to 100 Percentile
Sea Level Change	0 to 30cm	31to 60cm	61to90cm	91 to 120cm	120 to 150cm

2.4 Design of Scenarios

The InVEST model was employed to assess the potential impact of SLR on coastal risk exposure and the role of coastal habitats in mitigating this exposure. To explore different possibilities, four specific scenarios were developed (**Table 4**), each based on different assumptions regarding habitat conditions and the influence of SLR on the coastal system. These scenarios were chosen to provide a comprehensive understanding of how varying levels of habitat protection and different SLR projections may impact coastal exposure. The first scenario reflects the current state, with both habitat protection and present climatic conditions. The second scenario assumes habitat loss, resulting in the absence of protection, while maintaining the same climatic conditions. The third scenario explores future conditions under a projected SLR of 1.5 m by 2100, with habitat protection. The fourth scenario considers the same future conditions but assumes the absence of habitat protection. By comparing these scenarios, the model allows for an assessment of how habitat loss or conservation influences exposure to coastal hazards. This offers insight into the potential effects of varying levels of habitat protection under future climate conditions.

Table 4: Description of Scenarios Analyzing Habitat Protection and Sea-Level Rise

Scenario 1	a) This scenario analyzes current habitat protection and climate conditions. It assumes that coastal ecosystems continue to provide their natural protective functions against storm surges and erosion.
Scenario 2	b) This scenario illustrates the potential consequences of lacking habitat protection. It assumes that natural coastal ecosystems no longer act as protective barriers, resulting in the coastline being more exposed to coastal hazards. Climatic conditions will remain unchanged.
Scenario 3	c) This scenario projects to the SLR (1.5 m by 2100) while maintaining current habitat protection. It explores the future situation where coastal ecosystems continue to offer protection against coastal hazards despite the SLR.
Scenario 4	d) This scenario also presents projections for SLR changes (1.5 m by 2100) but assumes the loss of natural habitat protection. It simulates the situation

2.5 Spatial autocorrelation analysis

Spatial autocorrelation is a statistical measure that quantifies the similarity of geographic attributes or features among adjacent areas. It is a technique used to analyze the spatial clustering of values (similarity or dependence) (Ai *et al.*, 2022). This analysis aims to understand the variables influencing the spatial distribution of EI. Spatial autocorrelation can be calculated using several methods, such as the Moran Index (MI), Geary's C, or cross-correlation functions (Sajjad *et al.*, 2020).

The InVEST model exposure index output consists of points distributed along the shoreline, each corresponding to an EI value. To understand the logic of the distribution of points along the coastline, determine if each point and its adjacent points represented a similarity in the aggregation of the coastal EI. The spatial autocorrelation was assessed at two scales: Global Spatial Autocorrelation and Local Spatial Autocorrelation

The Global Moran Index (GMI) was used to assess the geospatial autocorrelation of EIs for the Sale coastline under the four scenarios adopted. The MI was calculated by Geoda software version 1.18, applying (Equation 2). The index always belongs to the interval between [-1, 1]; if the value is close to -1 or 1, this indicates high (negative/positive) autocorrelation, whereas if the value falls to 0, this indicates low autocorrelation (Anselin, 1995, 2016).

$$MI = \frac{n \sum_{i=1}^n \sum_{j=1}^n w_{ij} (X_i - \bar{X})(X_j - \bar{X})}{\sum_{i=1}^n \sum_{j=1}^n w_{ij} \sum_{i=1}^n (X_i - \bar{X})^2} \quad (\text{Equation 2})$$

Where n is the number of space observation objects in the study area; X_i and X_j are the values of the i -th and j -th observation objects at the spatial position, respectively; $\bar{X}$ is the average observation value of all objects; w_{ij} is the spatial weight matrix, representing the adjacency relationship between the i -th and j -th observation objects. If objects i and j are adjacent, $w_{ij} = 1$; if they are not adjacent, $w_{ij} = 0$. The value range of Moran's I is between 1 and -1.

The next step is to calculate the Local Indicators of Spatial Association (LISA) using the Local Moran Index (LMI) (Equation 3) (Ord and Getis, 1995). This step assesses the degree of spatial heterogeneity between specific zones and their adjacent areas.

$$LMI = \frac{\sum_{j=1}^n w_{ij} (X_i - \bar{X})(X_j - \bar{X})}{\sum_{i=1}^n (X_i - \bar{X})^2} \quad (\text{Equation 1})$$

3. Results

3.1 Distribution of Coastal EI of the model outputs

The model produces 267 points along the coastline with a 100 m resolution, capturing the degree of exposure to coastal hazards. Figure 5 map illustrates geographical variations in EI, while Figure 6 presents a complementary bar chart highlighting the proportional distribution of each risk class under four scenarios. The results highlight significant variations in exposure levels based on two critical factors: the presence or absence of habitat protection and the projected consequences of SLR by 2100.

In Scenario 1 (Figures 5(a) and 6), 51.3% of coastal segments show moderate exposure, mainly along the central coastline. High-risk areas cover 30.4%, with 15.4% classified as high risk and 15% as very high risk, with concentrations around Sidi Moussa and the Bouregreg estuary. Low-risk areas represented 18.32% (14.2% low risk, 4.12% very low risk), mainly in the northern and southern parts. These results underline the protective role of natural habitats under current conditions, helping to limit risk zones and balance exposure levels. In scenario 2 (Figures 5(b) and 6), where natural habitat protection is eliminated, coastal exposure increases significantly. High-risk areas now dominate 56.9% of the coastline, particularly near Bouknadel and Hay Chemaou, while very high-risk areas extend to 31.5%, particularly around Sidi Moussa and the Bouregreg estuary. Moderate-risk areas are reduced to 10.1%, confined to the central

segments, and low-risk areas disappear almost completely, dropping to just 1.5%. These results underline the crucial role of natural habitats in mitigating coastal exposure. Without them, moderate- and low-risk areas are transformed into high-risk zones, considerably increasing overall vulnerability. In Scenario 3 (**Figures 5(c) and 6**), which models SLR impacts while maintaining natural habitats, the risk distribution remains similar to Scenario 1. Moderate risk remains predominant (51.3%), while high- and very high-risk areas (30.4%) concentrate in SLR-vulnerable areas such as Sidi Moussa and the Bouregreg estuary. Low-risk sections decrease to 18.32%. This scenario reinforces the long-term value of habitat conservation in sustaining moderate exposure levels, even under climate change conditions. Finally, in Scenario 4 (**Figures 5(d) and 6**), the combined effects of SLR and habitat loss result in the most severe impacts. Nearly 44% of coastal segments shift to very high risk, while 39.3% fall into the high-risk category, indicating widespread exposure. Moderate-risk segments shrink to 16.9%, and low-risk areas vanish entirely. Compared to Scenario 3, high-risk areas expand significantly, and moderate-risk zones are almost eliminated. This scenario underscores the compounding dangers of habitat degradation and SLR, pushing nearly all coastal segments into the highest risk categories.

Overall, the results demonstrate that SLR amplifies coastal exposure to risk. However, conserving natural habitats remains an essential strategy to mitigate this exposure. Without protective ecosystems, risk levels increase sharply, particularly in scenarios involving SLR impacts.

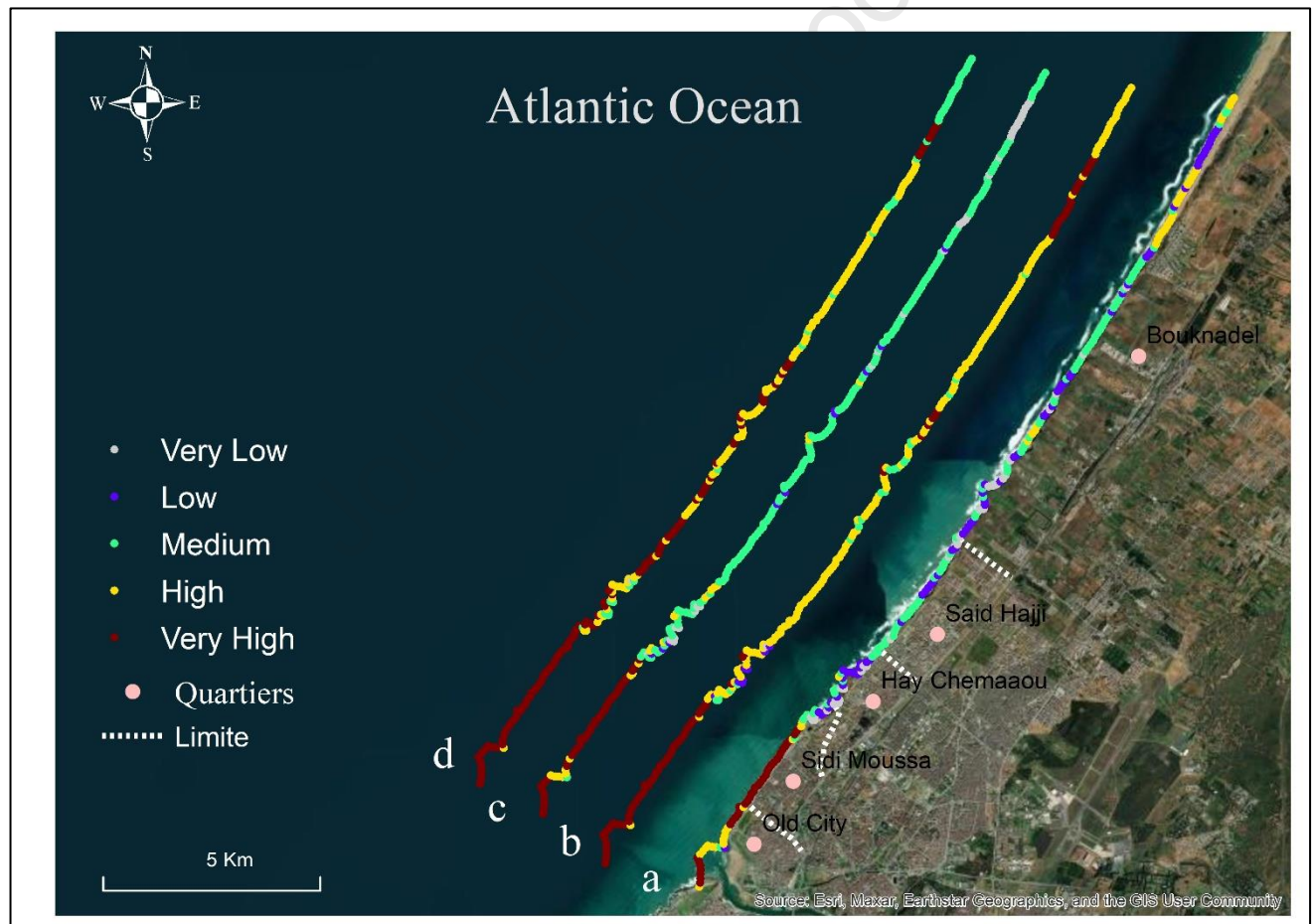


Figure 5: coastal EI by scenario; The change in color of the points along the shoreline represents the change in the degree of EI in space.

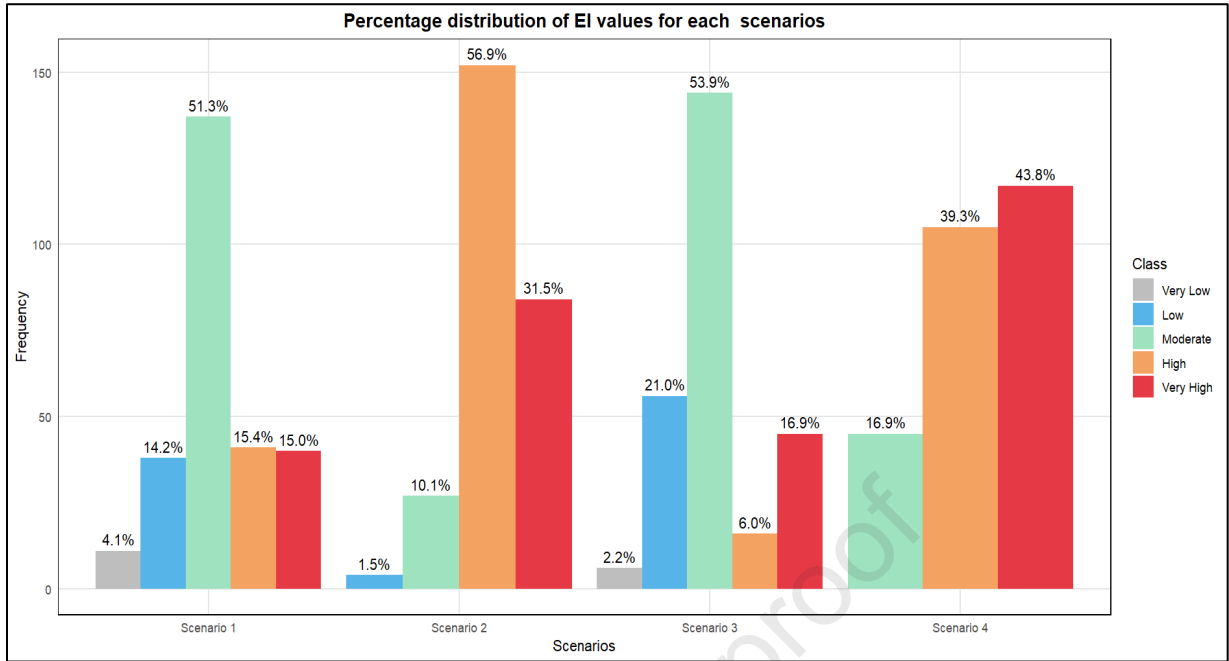


Figure 6: The histogram shows the EI values of each class for the different scenarios. The Y axis shows the calculated EI values, and the X axis shows the EI value classes in each scenario.

3.2 Geospatial autocorrelation analysis of EI

Figure 7; a, b, c, and d present a scatter plot that illustrates the relationship between the exposure variable and its lagging exposure, showing a dense distribution of points for all the scenarios. The MI values are positive and close to 1. The results show a similarly significant positive spatial autocorrelation, indicating that the calculated EIs have a high tendency to be identical to other geographically adjacent values (**Figure 7; a, b, c et d**). The Z and P-values confirm a statistically significant geospatial autocorrelation of the EI. The P-value of all scenarios, $0.001 < 0.5$, indicates that the index's similarity is statistically significant. The Z-value values are very far from 0 and above 2.58 (critical test), which presents the necessary value and passes the significance test of the exposure index $\alpha = 0.001$, representing a 99% confidence level of the index distribution (Table 5). This indicates that EIs along the coastline vary as a function of the geospatial and environmental factors associated with each unit.

Table 5: Global Moran's I static significance, and test of coastal exposure level coastline. It should be noted that we have considered pseudo-P-values based on 999 permutations (the most extreme that can be obtained).

Scenarios		Moran I	Z-value	P-value	Threshold $\alpha=0.01$
Current	Scenario 1	0.72	19.18	0.001	2.58
	Scenario 2	0.71	18.80	0.001	2.58
Future	Scenario 3	0.79	20.63	0.001	2.58
	Scenario 4	0.80	20.14	0.001	2.58

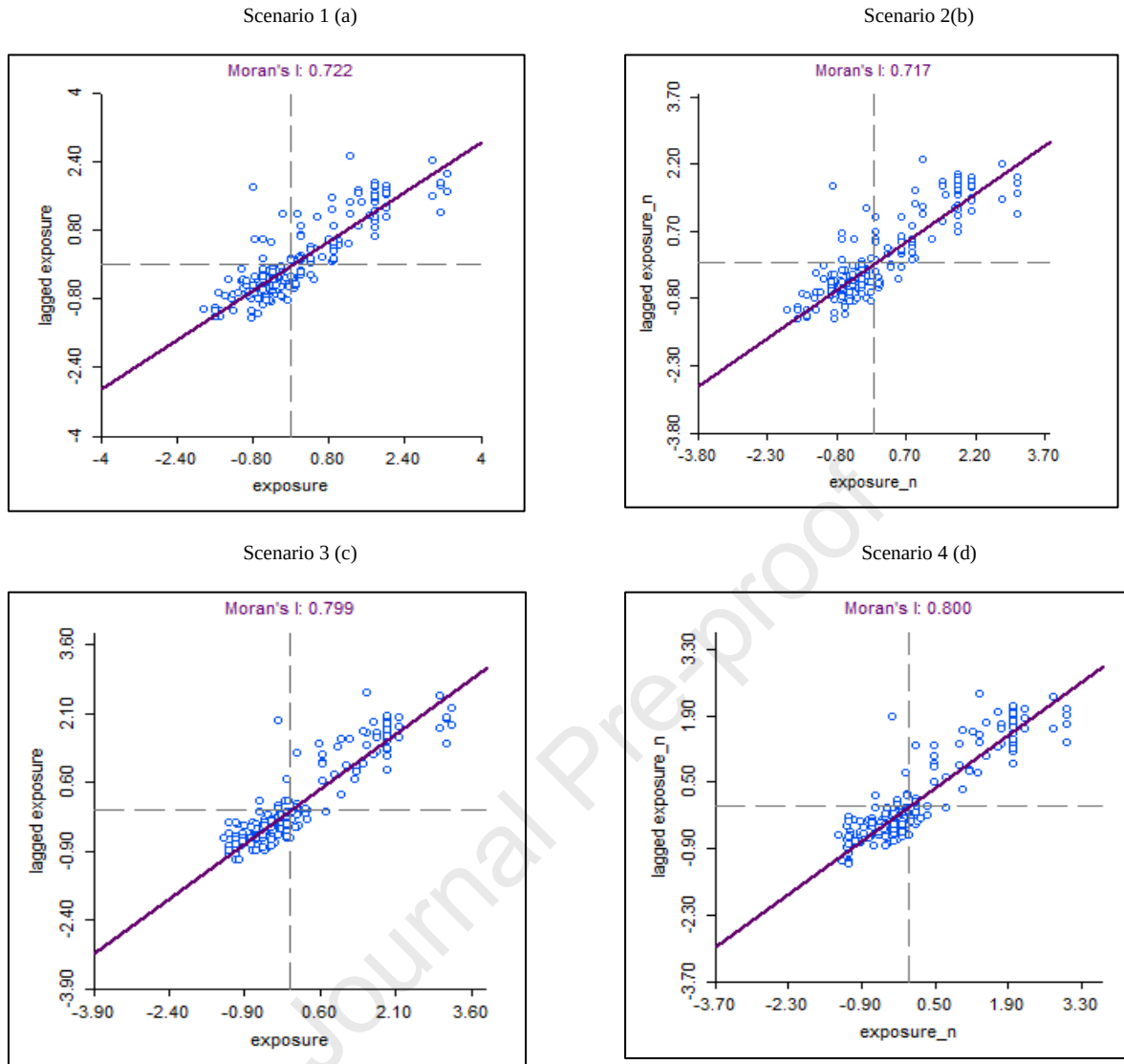


Figure 7: The Moran's I scatter plot of Coastal exposure by scenarios

The results of the LISA analysis are provided using cluster maps. The cluster map provides information on the spatial aggregation of the IE coastline (**Figure. 8**) and (**Table 6**). The table displays a LISA analysis of spatial clusters (H-H, L-L, H-L, L-H) for the four chosen situations. The findings indicate that H-H (Significant concentration in the high index exposure zone) and L-L (Significant concentration in the low index exposure zone) clustering are predominant, corresponding to 16.22% and 17.22% in scenarios 1 and 2. In projected scenarios, H-H experiences some increase, getting 18.72% (scenario 4), suggesting a concentration of high-value places, while L-L decreases in some instances, indicating an expansion of low values. Outliers (H-L and L-H) are few or absent, indicating minimal variation in these spatial associates.

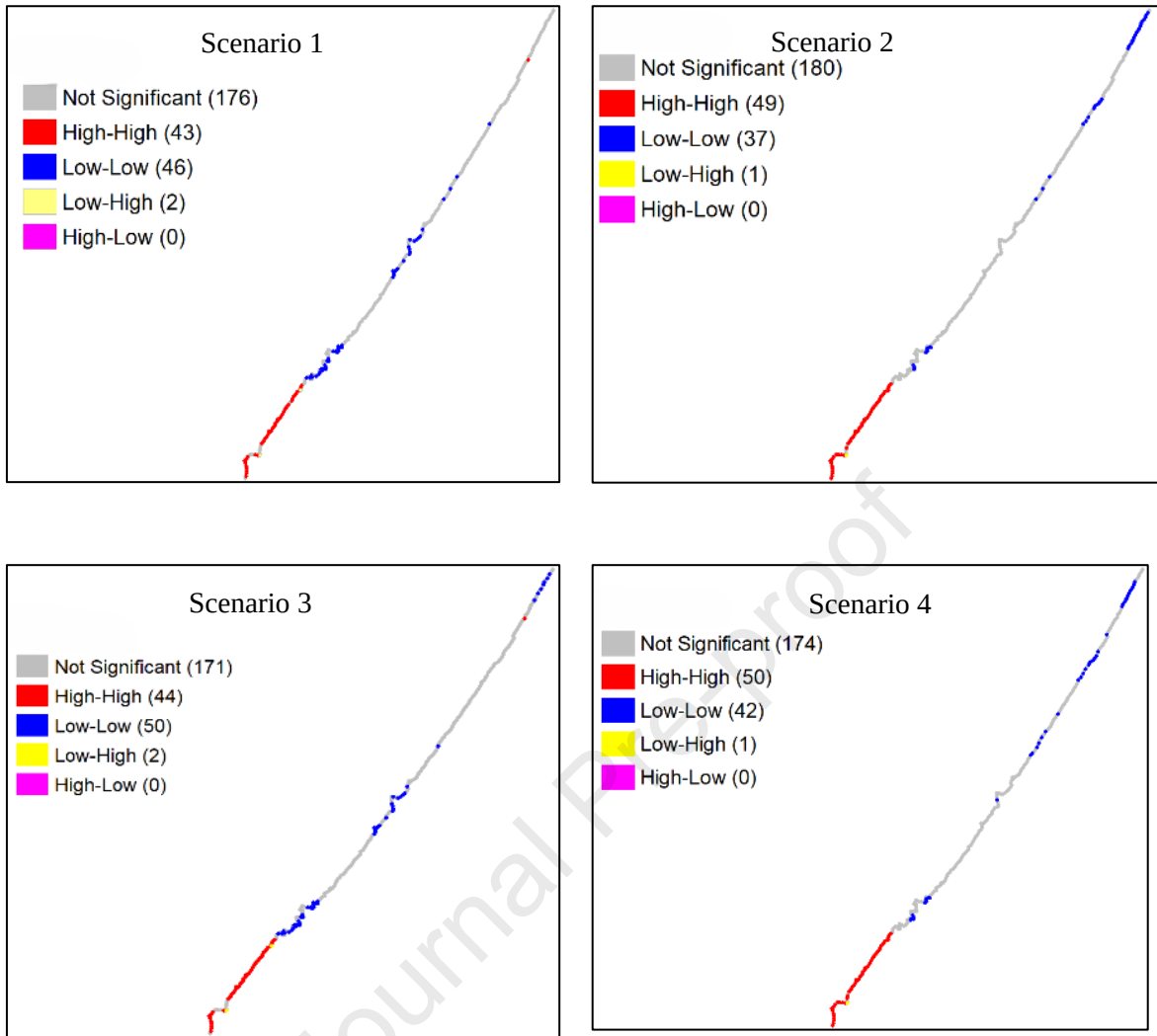


Figure 8: LISA cluster agglomeration maps of EI for each scenario.

Table 6: Type of local spatial autocorrelation of the EI in Sale coastline

Type	Current				Future			
	Scenario 1		Scenario 2		Scenario 3		Scenario 4	
	NB	%	NB	%	NB	%	NB	%
H-H	43	16.22	49	18.35	44	16.47	50	18.72
L-L	46	17.22	37	13.85	50	18.72	42	15.73
H-L	2	0.74	1	0	2	0.74	1	0
L-H	0	0	0	0	0	0	0	0
	176	65	180	67	171	64	174	65
Total	267	100	267	100	267	100	267	100

3.3 Field investigations

In the case of the first scenario, a thematic map was created to validate the result of the InVEST model and show the distribution of the vulnerability index in the coastal studied area (**Figure 9**). Also, several

georeferenced points and photos were added, and linked to the model observation. Three areas were considered, two with high exposure, and the third with a low exposure risk. These field investigations have enriched the analysis and are a tool for risk assessment and accurate evaluation of the results. The zone of the nations' beaches, sandy beach, is characterized by a vast expanse of sand, suggesting a relatively stable coastal zone in the immediate future, with a moderate to low exposure rate. The vast expanse of sand plays a key role in dissipating wave energy and absorbs some of the wave energy. Although the erosion process is less intense. The Sidi Moussa region shows a coastal cliff undergoing active erosion. This instability is visible in exposed rock strata and crevices. Infrastructure (Humans, buildings, and roads in the background) increases the region's vulnerability. This cliff appears high risk for coastal erosion, particularly in extreme climatic events or rising sea levels. These observations testify to the different vulnerability and dynamics of the coastline.



Figure 9: Field observation sites in Sale coastline

4. Discussion

The assessment of the exposure of the Sale coastline to coastal hazards was performed using a combined approach of a qualitative model and GIS database. This approach provides to identify the areas high risk and low protection areas. This study investigated spatial and temporal changes in Sale coastal exposure to

hazards measures over current and future (2100) periods. This section analyzes the spatial autocorrelation of (EI) along the shoreline. The Sale region is important for the projects (“*bk. EFH-MO-5 (2019)*”).

The results of this study integrate climate scenarios and biological data to explore hazard risks, focusing on the role of natural habitats in reducing the effects of coastal hazards. The SLR significantly increases the Sale coastline's exposure to coastal risks, it has boosted the area's exposure to very high risk (**Figure 5**). Coastal relief plays an important role in increasing the index of exposure to the risk of erosion and submersion caused by the hydrodynamics of the area and accentuated by the effect of climate change (*Arkema et al., 2013b; Jackson et al., 2020*). This effect becomes more pronounced as one moves further south, due to the differences in relief (altitude and bathymetry). The relatively low altitude of areas likely the estuary, Sidi Moussa, and Sidi Hajji results in increased exposure levels. In contrast, in Nation Beach high elevation decreases exposure. Under certain conditions, sea-level rise on a sandy segment can effectively reduce certain coastal risks, particularly those linked to dunes and beaches (Nation Beach) that can serve as natural buffer zones (*Ferreira, 2022*).

The ability of the habitat to protect itself depends on other factors such as geomorphology, location, condition, the dynamic and climatic forcing to which it is subjected, and its protection distance. Many studies have analysed the importance of ecosystems in protecting coastlines and have demonstrated their significant role in reducing risks. The Japanese have long practiced the concept of using coastal ecosystems to protect populations through what is known as grey infrastructure (*Kato and Huang, 2021*). One study showed the potential of coastal habitats to reduce coastal risks for coastal communities (*Al Ruheili and Boluwade, 2023*). This study has shown the importance and effectiveness of habitat in reducing the extent of coastal risk and the severity of damage caused by their loss. Regarding the role of natural habitat in mitigating SLR, the map has shown the potential of ecosystems to reduce the exposure of the Sale coastline because of the 2100 projection. It has highlighted specific locations where this ecosystem-based reduction is particular. A reduction in the abundance of habitat and SLR leads to an increase in the percentage of highly exposed areas (Hopper and Meixler, 2016; Jackson et al., 2020; Ballesteros and Esteves, 2021; Zhang et al., 2021; Aja et al., 2023). The EI reduction results demonstrate the importance of sand dunes (*Cabral et al., 2017*). Kelp, brown seaweed, is essential in protecting the coastline and reducing erosion. This type of seaweed has many benefits for the environmental cycle of wetlands, but also for protecting coastlines against the risk of erosion, by breaking the force of waves and stabilizing the ocean floor, thanks to the vast mats it forms. Unfortunately, they are also vulnerable to climate change, pollution, and habitat destruction (*Teagle et al., 2017*). Seagrass beds also offer protection to the coast and have reduced EI in the Sidi Taibi area. Some studies propose spatial planning and integrated management approaches as solutions to minimize the damage caused by such events (*Castellanos-Tapia et al., 2022; Cooley et al., 2022; Manes et al., 2023*). Research in Southern Brittany suggests the algobox® system, which uses seaweed-filled cages to stabilize sediments and protect dunes against erosion, while also including protection via natural ecosystems to increase coastal resilience (*Sedrati and Cochet, 2015*).

Coastal geomorphology is essential in influencing coastal exposure to hazards exacerbating or mitigating risk caused by waves, storms, and sea-level rise (SLR). The shape and characteristics of the coastline are critical in determining how natural forces interact with it (*Jackson et al., 2020*). The Nation's beaches benefit from protective formations such as sandy dunes; even with SLR, the area remains at a low exposure can be explained as SLR with sandy formations can form a protective formation (*Ferreira, 2022*). The Sidi Haji segment is protected by high cliffs that play an important role in attenuating wave energy, but exposure is augmented by SLR. The Sidi Moussa area and Bouregreg estuary are the most exposed regions, where the geomorphology, dominated by low cliffs, reflects an unfavorable configuration in the face of wave and storm impacts. The observation is in line with the findings of Cabral et al.(2017), according to which areas with unfavorable geomorphology are more exposed to coastal hazards. These contrasts underline the importance of adapted coastal management, including hybrid solutions combining sustainable infrastructure and ecosystem preservation (*Sutton-Grier et al., 2015*). These results highlight the diversity of coastal geomorphologies and their influence on vulnerability to marine hazards, requiring continuous monitoring to anticipate and reduce environmental and climatic impacts.

The continental shelf has a considerable influence on coastal exposure to hazards. Its extension between the continent and the ocean acts as a natural buffer. As seen at the national beach, a large plateau disperses wave energy, reducing the impact of waves and swell on the coastline. Conversely, areas such as Sidi

Moussa and said hajji, located closer to the ocean, are more exposed due to their proximity to the ocean (Ai *et al.*, 2022).

The spatial analysis of the autocorrelation of the EI configured by the global Moran index was evaluated based on a 99% confidence level, and the results showed a highly significant distribution of the EI over the coastline. Based on LISA results, we found a significant clustering of high EI in the southern part of the area (the Bouregreg estuary, Sidi Moussa, and said hajji) where there is a low elevation and a higher risk of SLR and where there are other units with similar hydrodynamic and physical conditions, the number of H-H increases with the disappearance of habitat protection. The north of the study area (Bouknadel and Sidi Taibi) has a concentration of low EI in areas of high altitude and particular geomorphology (sandy dunes and accentuated coastline). Other adjacent points with similar geomorphological profiles and hydrodynamic conditions surround it. The number L-L of EIs is declining as a result of habitat loss.

Field observations validated the model results by providing a more accurate vision of the coastal location. The areas identified as highly exposed to coastal hazards correspond to vulnerable sectors, notably the corniche and the Sale marina, which have considerable economic and tourism value, and Borj Adoumoue, which is important for tourism and history. Although these infrastructures are of considerable value, their presence in these vulnerable areas increases the risk of flooding and submersion, particularly as sea levels rise.

Comparison with other surveys worldwide

This study integrates climate scenarios and biological data to assess the influence of physical and ecological factors on coastal exposure. Our results align with international research while providing regional perspectives. Several studies (Corrales *et al.*, 2020; Jackson *et al.*, 2020; Al Ruheili and Boluwade, 2023) have highlighted the role of coastal and marine ecosystems in risk mitigation and enhancing ecosystem services. Indeed, these observations confirm that preserving natural habitats allows for maintaining moderate exposure levels, even in the presence of SLR (Reguero *et al.*, 2018). The integration of natural habitats as tools for CC adaptation has been explored in several studies (Depietri and McPhearson, 2017; Narayan *et al.*, 2017), highlighting their ability to absorb and recover from disturbances. The absence of coral reefs, for example, has been associated with increased coastal exposure, necessitating intensive management strategies to restore their resilience (Elliff and Kikuchi, 2017). Similarly, Cabral *et al.* (2017) demonstrated that highly exposed coastal areas often correspond to unfavorable morphologies, particularly in low altitude regions. These results confirm this relationship, highlighting that habitat destruction exacerbates high-risk areas. In parallel, although some research focuses on gray infrastructure (Arkema *et al.*, 2017; Kato and Huang, 2021), this study emphasizes the complementary role of these infrastructures to natural ecosystems in risk reduction. Other approaches, such as spatial planning and integrated coastal zone management, are also suggested to minimize the impacts of extreme events (de Vrees, 2021). Ultimately, these findings align with global trends established by Ai *et al.* (2022) and Zhang *et al.* (2021), which associate habitat loss and SLR with increased exposure. These conclusions reaffirm the importance of conserving natural habitats to strengthen resilience against future climate scenarios.

The limitations of the model

The method has many limitations (Mukhopadhyay *et al.*, 2024). The wave and wind data are obtained from the InVEST database, which is very efficient for global simulations (Onat *et al.*, 2018; Shao *et al.*, 2018). However, it does not consider all possible details and variations, particularly the dynamic nature of the study area and the scenarios proposed for future projections. The model oversimplifies the complexity of dynamic interactions. This simplification does not fully reflect the complex reality of coastal processes, which limits the precision or potential validity of predictions based on this model (Sajjad *et al.*, 2020). Moreover, the dynamics associated with extreme storms are complex and can give unexpected scenarios (Silver *et al.*, 2019). It cannot perform a quantitative assessment of erosion or flooding risks. Habitat rankings demonstrate variations in their protective capabilities based on a literature assessment. Therefore, precise processes and significant empirical confirmation are needed (Wolff *et al.*, 2016). This model is recommended due to its input and output data transparency, allowing for quick testing of various climate scenarios (Arkema *et al.*, 2013b, 2017; Hopper and Meixler, 2016; Cabral *et al.*, 2017). Despite certain limitations and gaps in its ability to prevent areas at risk, the research shows a certain reliability and good match

between the exposure index, as estimated by the coastal vulnerability model, through its validation by empirical data based on real observations of the impact of coastal hazards. In the United States (Arkema et al., 2013) have reinforced their results by comparing them with actual hazard death data for each unit, which are obtained from the spatial database of hazard events and losses for the unit condition. In Mozambique (Cabral et al., 2017) and in East Africa, (Ballesteros and Esteves, 2021) confirmed the model's effectiveness using a database of losses and damages caused by geological causes linked to DesInventar meteorological conditions. The most exposed locations have experienced more catastrophes and losses than previous data from the DesInventar database. Others have completed their study with static analysis (Hopper and Meixler, 2016; Onat et al., 2018; Cunha et al., 2021; Gammon et al., 2023) and spatial autocorrelation of the distribution of the EI (Ai et al., 2022) the same for this current research.

Recommendations and Importance of the Study for Authorities

This study provides valuable coastal planning and management guidelines, identifying areas at risk from SLR and highlighting the importance of natural habitats in mitigating coastal hazards, examining current and projected risks. The study offers a solid foundation for developing resilience strategies that can help authorities adapt to the effects of climate change.

Conservation of natural habitats: Prioritize preserving and restoring ecosystems like sand dunes, seagrass, and mangroves. These habitats play an essential role in reducing the impact of coastal hazards by acting as natural buffers against flooding and erosion.

Implement adaptive management strategies: Adopt flexible, responsive approaches to adapt to climate change and its associated risks. In particular, this means constantly monitoring the coastal zones most at risk from SLR, and integrating the latest scientific information into decision-making processes.

Combining natural ecosystems and sustainable infrastructure to reinforce coastal resilience. For example, integrating innovations such as **algobox®** systems can help stabilize dunes and protect coasts while preserving ecological integrity.

Conclusion

This study aimed to develop a qualitative analysis of the Sale coast and assess its exposure to coastal hazards, particularly sea-level rise, using an index-based model encompassing several biophysical and climatic parameters and factors. The research demonstrated the increase in exposure in the future and identified the importance of natural habitat in maintaining coastal protection and assessing the extent of coastal exposure. This research had two objectives: the first was to analyze the distribution of the EI as a function of the selected variables, and the second was to examine the aggregation of this distribution using the Moran index (global and local). The results took the form of exposure maps, showing that the hottest parts at risk are located south of the coastline. In contrast, the northern part of the zone is less risky due to geomorphology, altitudinal relief, and the location of habitat protection, with a logical distribution of the EI by a significant positive correlation. The main objective of these results was to provide decision-makers with an overview of the fragility and resilience of the Sale coast in its current state under current climate conditions and in the long term through CC-SLR risk assessments and to support efforts invested in the region, particularly in the eastern part of bouregreg. These results could be useful to those innovating in coastal protection and integrated coastal zone management to protect and reduce the losses and damage associated with humid zones. InVEST modeling has a twofold benefit: this approach has clearly shown the fragile points of the coastline exposed to coastal risk and SLR, but it has equally documented the importance of habitat and grey infrastructure as a useful solution for reducing risk along the Sale coastline. This methodology and its results are, therefore, very useful for facilitating the assessment of this risk in other areas. They also offer solutions that promote sustainability and encourage the adaptive management of wetland ecosystems. This study gives a preliminary analysis of the fragility and resilience of the coastline, and an understanding of how it is affected by climatic factors, paving the way for further, more in-depth studies in the region, and inspiring new research that can complement these results by combining other methods or other types of data to complete the missing details. Surveys using methods that uniquely and accurately quantify each watercourse, as well as a better knowledge of the region's geology and an understanding of how the geomorphology of the coast can be mitigated with this risk. The study remains a solid basis for future work, a rich and concrete path that can be exploited in other appropriate fields or developed to meet

similar needs and challenges

CRediT authorship contribution statement

Samya Lamhadri: Conceptualization of this study, Methodology, Software, Formal analysis, Data curation, Writing, Original draft preparation, visualization, Investigation, and Funding acquisition. **Nadia Senechal:** methodology, validation, supervision. **Abdessalam Ouallali:** methodology, validation, supervision. **Mohammed El Hafyani:** validation, Review. **Driss Chahid:** validation, Review. **Mohammed Karim Benhachmi:** methodology, validation, Project administration, validation, supervision, Review.

Declaration of Competing Interest

The authors declare that there are no conflicts of interest regarding the publication of this research

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Declaration of Interest Statement

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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